

Currency: All costs in US Dollar (\$) | Exchange Rate: 1 USD = ₹84.00 | Report Date: 13 January 2024

#### PATIENT INFORMATION

**Name:** Rahul Sharma  
**Patient ID:** PT-2024-00123  
**DOB:** 15 May 1990 | **Age:** 33 Yrs | **Gender:** Male  
**Blood Group:** B+ | **Weight:** 72 kg | **Height:** 175 cm  
**Ward:** Ortho ICU – Room 204  
**Fasting Status:** Non-Fasting (Emergency)

#### LAB & REFERRAL INFO

**Referred By:** Dr. Anil Kapoor (MS – Ortho)  
**Pathologist:** Dr. Sunita Rao (MD – Pathology)  
**Lab Technician:** Mr. Vijay Patil (DMLT)  
**Sample Type:** Venous Blood + Urine (Mid-stream)  
**Priority:** **STAT – Emergency**  
**Total Lab Cost:** **\$261.90**

#### CRITICAL VALUE ALERT – Physician Notified at 10:45 AM

● **Troponin I:** 4.82 ng/mL (Ref: 0.00–0.04) – Critical High ↑↑ – AMI Confirmed | ● **D-Dimer:** 3.8 µg/mL (Ref: 0.0–0.5) – Critical High ↑↑ – Active Thrombosis | ● **Hb:** 8.2 g/dL – Low ↓ – Acute Anemia | ● **WBC:** 14,500/µL – High ↑ – Infection/Stress Response

#### LAB SUMMARY

6

Test Panels

40

Total Tests

22

Normal

12

Abnormal

6

Critical

**\$261.90**

Total Cost

#### ♦ PANEL 1: COMPLETE BLOOD COUNT (CBC) – \$26.19

| #                      | Test Name               | Result   | Unit       | Normal Range      | Flag     | Result  |
|------------------------|-------------------------|----------|------------|-------------------|----------|---------|
| 1                      | Hemoglobin (Hb)         | 8.2      | g/dL       | 13.0–17.0         | ↓ LOW    | Low     |
| 2                      | White Blood Cells (WBC) | 14,500   | /µL        | 4,500–11,000      | ↑ HIGH   | High    |
| 3                      | Red Blood Cells (RBC)   | 3.8      | million/µL | 4.5–5.9           | ↓ LOW    | Low     |
| 4                      | Platelets               | 2,10,000 | /µL        | 1,50,000–4,00,000 | ✓ NORMAL | Normal  |
| 5                      | Hematocrit (HCT)        | 26.4     | %          | 40–52             | ↓ LOW    | Low     |
| 6                      | MCV                     | 82       | fL         | 80–100            | ✓ NORMAL | Normal  |
| 7                      | MCH                     | 27       | pg         | 27–33             | ✓ NORMAL | Normal  |
| 8                      | MCHC                    | 31.0     | g/dL       | 31.5–35.5         | ↓ LOW    | Low     |
| 9                      | Neutrophils             | 78       | %          | 50–70             | ↑ HIGH   | High    |
| 10                     | Lymphocytes             | 16       | %          | 20–40             | ↓ LOW    | Low     |
| 11                     | Monocytes               | 5        | %          | 2–8               | ✓ NORMAL | Normal  |
| 12                     | ESR                     | 45       | mm/hr      | 0–20              | ↑ HIGH   | High    |
| CBC – 12 Tests   Cost: |                         |          |            |                   |          | \$26.19 |

#### ♥ PANEL 2: CARDIAC MARKERS – \$59.52

| #                         | Test Name                | Result | Unit  | Normal Range | Flag        | Result   |
|---------------------------|--------------------------|--------|-------|--------------|-------------|----------|
| 1                         | Troponin I               | 4.82   | ng/mL | 0.00–0.04    | ↑↑ CRITICAL | Critical |
| 2                         | Troponin T               | 0.98   | ng/mL | 0.00–0.01    | ↑↑ CRITICAL | Critical |
| 3                         | CK-MB                    | 86     | U/L   | 0–25         | ↑ HIGH      | High     |
| 4                         | D-Dimer                  | 3.8    | µg/mL | 0.0–0.5      | ↑↑ CRITICAL | Critical |
| 5                         | CRP (C-Reactive Protein) | 48.2   | mg/L  | 0.0–5.0      | ↑ HIGH      | High     |
| 6                         | BNP                      | 310    | pg/mL | 0–100        | ↑ HIGH      | High     |
| 7                         | Prothrombin Time (PT)    | 13.5   | sec   | 11.0–13.5    | ✓ NORMAL    | Normal   |
| 8                         | INR                      | 1.1    | –     | 0.8–1.2      | ✓ NORMAL    | Normal   |
| Cardiac – 8 Tests   Cost: |                          |        |       |              |             | \$59.52  |

#### □ PANEL 3: LIVER FUNCTION TEST (LFT) – \$26.19

| #                     | Test Name                  | Result | Unit  | Normal Range | Flag     | Result  |
|-----------------------|----------------------------|--------|-------|--------------|----------|---------|
| 1                     | Total Bilirubin            | 0.9    | mg/dL | 0.2–1.2      | ✓ NORMAL | Normal  |
| 2                     | Direct Bilirubin           | 0.3    | mg/dL | 0.0–0.3      | ✓ NORMAL | Normal  |
| 3                     | SGOT (AST)                 | 42     | U/L   | 10–40        | ↑ HIGH   | High    |
| 4                     | SGPT (ALT)                 | 38     | U/L   | 7–56         | ✓ NORMAL | Normal  |
| 5                     | Alkaline Phosphatase (ALP) | 88     | U/L   | 44–147       | ✓ NORMAL | Normal  |
| 6                     | Total Protein              | 6.2    | g/dL  | 6.0–8.3      | ✓ NORMAL | Normal  |
| 7                     | Albumin                    | 3.4    | g/dL  | 3.5–5.0      | ↓ LOW    | Low     |
| 8                     | Globulin                   | 2.8    | g/dL  | 2.3–3.5      | ✓ NORMAL | Normal  |
| LFT – 8 Tests   Cost: |                            |        |       |              |          | \$26.19 |

#### □ PANEL 4: RENAL FUNCTION TEST (RFT) – \$23.81

| #                     | Test Name                 | Result | Unit   | Normal Range | Flag     | Result  |
|-----------------------|---------------------------|--------|--------|--------------|----------|---------|
| 1                     | Blood Urea Nitrogen (BUN) | 18     | mg/dL  | 7–25         | ✓ NORMAL | Normal  |
| 2                     | Serum Creatinine          | 1.1    | mg/dL  | 0.7–1.3      | ✓ NORMAL | Normal  |
| 3                     | eGFR                      | 95     | mL/min | >90          | ✓ NORMAL | Normal  |
| 4                     | Serum Sodium (Na+)        | 138    | mEq/L  | 136–145      | ✓ NORMAL | Normal  |
| 5                     | Serum Potassium (K+)      | 3.4    | mEq/L  | 3.5–5.0      | ↓ LOW    | Low     |
| 6                     | Serum Chloride (Cl-)      | 101    | mEq/L  | 98–106       | ✓ NORMAL | Normal  |
| 7                     | Uric Acid                 | 5.8    | mg/dL  | 3.5–7.2      | ✓ NORMAL | Normal  |
| RFT – 7 Tests   Cost: |                           |        |        |              |          | \$23.81 |

#### 🔴 PANEL 5: BLOOD SUGAR & LIPID PROFILE – \$19.05

| #                               | Test Name                | Result | Unit  | Normal Range | Flag     | Result  |
|---------------------------------|--------------------------|--------|-------|--------------|----------|---------|
| 1                               | Random Blood Sugar (RBS) | 168    | mg/dL | 70–140       | ↑ HIGH   | High    |
| 2                               | HbA1c                    | 6.2    | %     | <5.7         | ↑ HIGH   | High    |
| 3                               | Total Cholesterol        | 198    | mg/dL | <200         | ✓ NORMAL | Normal  |
| 4                               | LDL Cholesterol          | 128    | mg/dL | <100         | ↑ HIGH   | High    |
| 5                               | HDL Cholesterol          | 38     | mg/dL | >40          | ↓ LOW    | Low     |
| 6                               | Triglycerides            | 186    | mg/dL | <150         | ↑ HIGH   | High    |
| 7                               | VLDL                     | 37     | mg/dL | <30          | ↑ HIGH   | High    |
| Sugar & Lipid – 7 Tests   Cost: |                          |        |       |              |          | \$19.05 |

 **PANEL 6: URINE ROUTINE & MICROSCOPY – \$11.90**

| #                       | Test Name        | Result   | Unit | Normal Range       | Flag     | Result  |
|-------------------------|------------------|----------|------|--------------------|----------|---------|
| 1                       | Colour           | Yellow   | –    | Yellow/Pale Yellow | ✓ NORMAL | Normal  |
| 2                       | Clarity          | Clear    | –    | Clear              | ✓ NORMAL | Normal  |
| 3                       | pH               | 6.0      | –    | 4.5–8.0            | ✓ NORMAL | Normal  |
| 4                       | Specific Gravity | 1.018    | –    | 1.005–1.030        | ✓ NORMAL | Normal  |
| 5                       | Protein          | Trace    | –    | Negative           | ↑ HIGH   | High    |
| 6                       | Glucose          | Negative | –    | Negative           | ✓ NORMAL | Normal  |
| 7                       | RBC (Microscopy) | 2–4      | /HPF | 0–2                | ↑ HIGH   | High    |
| 8                       | WBC (Microscopy) | 1–2      | /HPF | 0–5                | ✓ NORMAL | Normal  |
| Urine – 8 Tests   Cost: |                  |          |      |                    |          | \$11.90 |

 **Lab Cost Summary (USD \$)**

|                               |                |
|-------------------------------|----------------|
| CBC – 12 Tests                | \$26.19        |
| Cardiac Markers – 8 Tests     | \$59.52        |
| LFT – 8 Tests                 | \$26.19        |
| RFT – 7 Tests                 | \$23.81        |
| Blood Sugar & Lipid – 7 Tests | \$19.05        |
| Urine Routine – 8 Tests       | \$11.90        |
| Insurance Covered             | – \$150.00     |
| Patient Paid                  | – \$61.90      |
| <b>Outstanding Balance</b>    | <b>\$50.00</b> |

 **Results Overview**

|                       |                 |
|-----------------------|-----------------|
| Total Panels          | 6               |
| Total Tests Done      | 40              |
| Normal Results        | 22              |
| Abnormal Results      | 12              |
| Critical Values       | 6               |
| <b>Total Lab Cost</b> | <b>\$261.90</b> |

 **LAB PROCESSING TIMELINE**

13 Jan 2024 – 06:00 AM

**Sample Collection**

Venous blood (5 vials) and mid-stream urine collected by Mr. Vijay Patil. STAT priority marked. Samples sent to lab immediately.

13 Jan 2024 – 06:30 AM

**Lab Processing Started**

All 6 panels processed simultaneously. CBC processed on Sysmex XN-1000 analyzer. Cardiac markers on Troponin-specific ELISA assay.

13 Jan 2024 – 10:30 AM

**Results Generated & Verified**

All 40 tests completed and verified by Dr. Sunita Rao. Critical values flagged automatically. Report generated and released.

13 Jan 2024 – 10:45 AM

**Critical Values Reported to Physician**

Dr. Anil Kapoor notified of Troponin I (4.82 ng/mL) and D-Dimer (3.8 µg/mL) critical values. Immediate cardiology intervention initiated.

 **PATHOLOGIST'S INTERPRETATION & NOTES**

**Dr. Sunita Rao – MD Pathology, NABL Verified:**

**CBC:** Anemia (Hb 8.2 g/dL) with low RBC and HCT consistent with acute blood loss post-RTA. Elevated WBC (14,500) with neutrophilia (78%) — acute inflammatory/stress response. ESR elevated. Platelet count normal — no bleeding disorder. Iron supplementation recommended post-stabilization.

**Cardiac Markers:** Troponin I critically elevated (4.82 ng/mL) — confirms Acute Myocardial Infarction (AMI). Troponin T also elevated. D-Dimer elevation (3.8 µg/mL) indicates active thrombotic process — DVT/PE risk. CRP elevation confirms systemic inflammation. BNP elevated — mild cardiac stress. STAT cardiological intervention warranted.

**LFT:** SGOT mildly elevated (42 U/L) — likely secondary to muscle trauma rather than primary hepatic injury. Other LFT parameters within normal limits. Low albumin — acute phase response. Monitor LFT in follow-up.

**RFT:** Renal function well preserved (eGFR 95 mL/min, Creatinine 1.1). Mild hypokalemia (K<sup>+</sup> 3.4 mEq/L) — oral/IV potassium supplementation recommended. All other electrolytes normal.

**Metabolic:** Elevated RBS (168 mg/dL) with borderline HbA1c (6.2%) — stress-induced hyperglycemia vs pre-diabetic state. LDL elevated (128 mg/dL) with low HDL (38) — dyslipidemia contributing to cardiovascular risk. Statin therapy to be considered post-recovery. VLDL mildly elevated.

**Urine:** Trace protein and 2–4 RBC/HPF — likely secondary to trauma/stress. No significant pathology. Repeat urine analysis recommended after 72 hours.

**Follow-up Labs Recommended:** Repeat Troponin at 6h and 12h | Repeat CBC after 48h | Fasting Lipid Profile post-discharge | HbA1c recheck in 3 months.